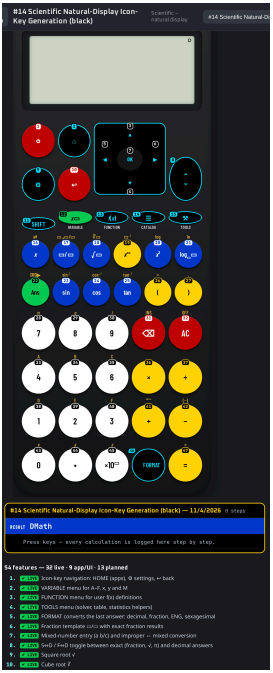
U-45 CALCULATORS

User Manual

Calculator #14 Scientific Natural-Display Icon-Key Generation (black)

Scientific – natural display



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#14 — Scientific Natural-Display Icon-Key Generation (black)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example  . Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic (type the expression as written, then = / EXE).

Chapters

1. Getting started
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4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
10. Appendix A — Drill answers
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12. Appendix C — Glossary

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1 · Getting started

Open U-45 CALCULATORS, choose **#14** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#28, #29, #30, #33, #34, #35, #38, #39, #40, #43, #44, #45
op	Operators & execute	#7, #19, #26, #27, #36, #37, #41, #42, #47
fn	Functions	#16, #17, #18, #20, #21, #23, #24, #25
mem	Memory, variables & registers	#12, #22
clr	Clear, delete & power	#1, #10, #31, #32
mod	Shift, modes, menus & navigation	#2, #3, #4, #5, #6, #8, #9, #11, #13, #14, #15, #46

2 · Features

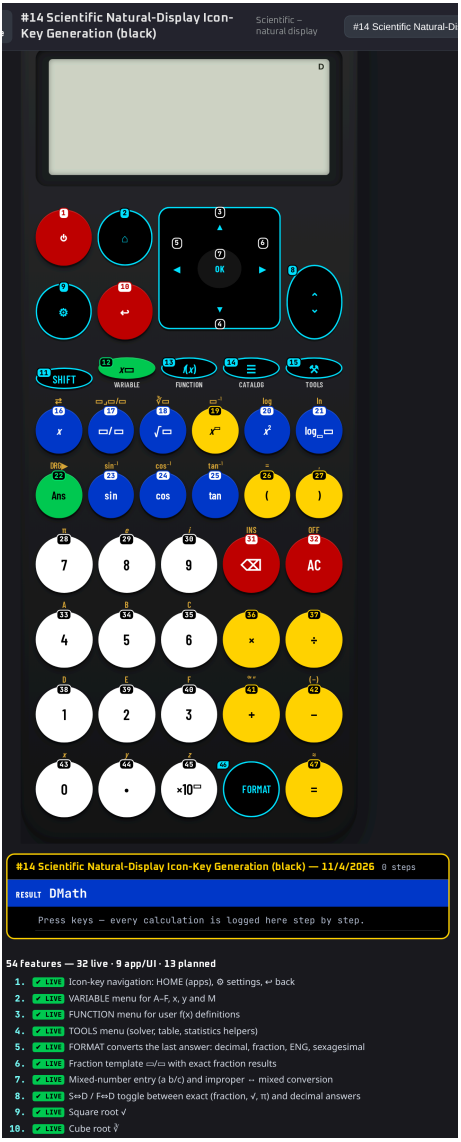
54 features: 32 live on the keys, 9 app/UI features, 13 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Icon-key navigation: HOME (apps), ⚙ settings, ↩ back**
2. LIVE **VARIABLE menu for A–F, x, y and M**
3. LIVE **FUNCTION menu for user f(x) definitions**
4. LIVE **TOOLS menu (solver, table, statistics helpers)**
5. LIVE **FORMAT converts the last answer: decimal, fraction, ENG, sexagesimal**
6. LIVE **Fraction template ▭/▭ with exact fraction results**
7. LIVE **Mixed-number entry (a b/c) and improper ↔ mixed conversion**
8. LIVE **S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers**
9. LIVE **Square root √**
10. LIVE **Cube root $\sqrt[3]{}$**
11. LIVE **Square x^2**
12. LIVE **Reciprocal x^{-1} (1/x)**
13. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
14. LIVE **Common logarithm $\log_{10}$**
15. LIVE **Natural logarithm ln**
16. LIVE **Logarithm to any base $\log_a b$**
17. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
18. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**
19. LIVE **Angle units DEG / RAD / GRA with unit-suffix conversion (°, ʳ, ᵍ)**

20. LIVE Sexagesimal ° ′ ″ entry and decimal ↔ DMS conversion
21. LIVE π constant to full internal precision
22. LIVE Euler e constant
23. LIVE Scientific-notation entry ×10ˣ / EXP / EEX
24. LIVE Sign change (−) / +/− / CHS
25. LIVE Ans recall of the last result; operators auto-insert Ans
26. LIVE Named variables (A–F, X, Y, M … or A–Z) usable inside expressions
27. LIVE Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular
28. LIVE MODE / SETUP: Fix / Sci / Norm display, angle unit, fraction format
29. LIVE Backspace / DEL editing of the current entry
30. LIVE INS insert mode for editing in the middle of an expression
31. LIVE Replay ▲ ▼ — scroll back to previous calculations and re-edit them
32. LIVE CATALOG — alphabetical list of every function
33. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
34. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
35. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
36. APP/UI Show / hide key numbers with one button (remembered between sessions)
37. APP/UI Copy the current display text to the clipboard
38. APP/UI Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
39. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen

40. APP/UI **Reference photo viewer and tool-tip name on every key**
41. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
42. PLANNED **Spreadsheet app**
43. PLANNED **Distribution app**
44. PLANNED **Algorithm (flowchart) app**
45. PLANNED **Math box: dice / coin simulation**
46. PLANNED **QR-code export**
47. PLANNED **Ratio app**
48. PLANNED **Inequality app**
49. PLANNED **Prime factorisation**
50. PLANNED **Exam mode (locks memory & shows indicator)**
51. PLANNED **Answer history export to text**
52. PLANNED **Speech read-out of results**
53. PLANNED **Colour-blind safe key theme**
54. PLANNED **Custom shortcut keys**

3 · Numbered key map



Calculator #14 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	⏻	·	·	·	clr
2	⌂	·	·	·	mod
3	▲	·	·	·	mod
4	▼	·	·	·	mod
5	◀	·	·	·	mod
6	▶	·	·	·	mod
7	OK	·	·	·	op
8	^ ⌄	·	·	·	mod
9	⚙	·	·	·	mod
10	↩	·	·	·	clr
11	SHIFT	·	·	·	mod
12	x▭	·	·	·	mem
13	f(x)	·	·	·	mod
14	☰	·	·	·	mod
15	⚒	·	·	·	mod
16	x	⇄	·	·	fn
17	▭/▭	▭␣▭/▭	·	·	fn
18	√▭	∛▭	·	·	fn
19	x▭	$▭^{-1}$	·	·	op
20	x^2	log	·	·	fn
21	log▭▭	ln	·	·	fn
22	Ans	DRG▶	·	·	mem
23	sin	sin^{-1}	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	cos	$\cos^{-1}$	·	·	fn
25	tan	$\tan^{-1}$	·	·	fn
26	(	=	·	·	op
27	)	,	·	·	op
28	7	π	·	·	num
29	8	e	·	·	num
30	9	i	·	·	num
31	⌫	INS	·	·	clr
32	AC	OFF	·	·	clr
33	4	A	·	·	num
34	5	B	·	·	num
35	6	C	·	·	num
36	×	·	·	·	op
37	÷	·	·	·	op
38	1	D	·	·	num
39	2	E	·	·	num
40	3	F	·	·	num
41	+	°′ ″	·	·	op
42	−	(−)	·	·	op
43	0	x	·	·	num
44	•	y	·	·	num
45	×10▭	z	·	·	num
46	FORMAT	·	·	·	mod
47	=	≈	·	·	op

4 · Key-by-key reference

Every key on calculator #14, in number order, with every function it carries in this app.

#1 ⏻ `Clear, delete & power`

Primary: ON — power on

#2 ⌂ `Shift, modes, menus & navigation`

Primary: HOME — return to the main calculation screen

#3 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#4 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#5 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#6 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#7 OK `Operators & execute`

Primary: OK — confirm the highlighted item

#8 ⌃ ⌄ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

Keyboard: `ArrowUp`

#9 ⚙ `Shift, modes, menus & navigation`

Primary: Settings

#10 ↩ `Clear, delete & power`

Primary: Back ↩ — previous screen

#11 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#12 x▭ `Memory, variables & registers`

Primary: Variable list (store / recall)

#13 f(x) `Shift, modes, menus & navigation`

Primary: FUNCTION menu

#14 ☰ `Shift, modes, menus & navigation`

Primary: CATALOG — list of all commands

#15 ⚒ `Shift, modes, menus & navigation`

Primary: TOOLS — tools menu

#16 x `Functions`

Primary: Variable x for formulas, tables and solving

shifted (2nd / f / SHIFT layer) (press #11 first): ⇄ — S⇔D — toggle the result between exact (fraction, √, π) and decimal

#17 ▭/▭ `Functions`

Primary: Fraction template ▭/▭ (numerator over denominator)

shifted (2nd / f / SHIFT layer) (press #11 first): ▭⌟▭/▭ — Mixed-number template (whole, numerator, denominator)

#18 √▭ `Functions`

Primary: Square root √

shifted (2nd / f / SHIFT layer) (press #11 first): ∛▭ — Cube root ∛

Keyboard: r

#19 x▭ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #11 first): $▭^{-1}$ — Reciprocal x^{-1} (1/x)

Keyboard: ^

#20 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #11 first): log — Common logarithm $\log_{10}$

Keyboard: q

#21 log▭▭ Functions

Primary: log with any base: $\log_a(b)$

shifted (2nd / f / SHIFT layer) (press #11 first): ln — Natural logarithm ln

#22 Ans Memory, variables & registers

Primary: Ans — recall the last answer into the current expression

shifted (2nd / f / SHIFT layer) (press #11 first): DRG▶ — DRG — choose/convert the angle unit (Deg / Rad / Grad)

#23 sin Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #11 first): $\sin^{-1}$ — Inverse sine $\sin^{-1}$

Keyboard: s

#24 cos Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #11 first): $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

Keyboard: c

#25 tan Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #11 first): $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

Keyboard: t

#26 (Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #11 first): = — Equals — complete the calculation

Keyboard: (

#27) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

shifted (2nd / f / SHIFT layer) (press #11 first): , — Comma / argument separator (between function arguments, list items or coordinates)

Keyboard:)

#28 **7** `Digits & number entry`

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #11 first): π — π (3.14159265…) as an exact constant

Keyboard: 7

#29 **8** `Digits & number entry`

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #11 first): e — Euler's number e (2.71828…)

Keyboard: 8

#30 **9** `Digits & number entry`

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #11 first): i — Imaginary unit i (complex numbers)

Keyboard: 9

#31 ⌫ `Clear, delete & power`

Primary: DEL — delete the character at the cursor

shifted (2nd / f / SHIFT layer) (press #11 first): INS — INS — insert mode at the cursor

Keyboard: `Backspace` `Delete`

#32 AC Clear, delete & power

Primary: ON / AC — power on, or clear all

shifted (2nd / f / SHIFT layer) (press #11 first): OFF — OFF — power off (state is lost; press ON to restart)

#33 4 Digits & number entry

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #11 first): A — Variable A (store with STO/ →, recall by name)

Keyboard: 4

#34 5 Digits & number entry

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #11 first): B — Variable B (store with STO/ →, recall by name)

Keyboard: 5

#35 6 Digits & number entry

Primary: Digit 6

shifted (2nd / f / SHIFT layer) (press #11 first): C — Variable C (store with STO/ →, recall by name)

Keyboard: 6

#36 × `Operators & execute`

Primary: Multiplication ×

Keyboard: [*] [x]

#37 ÷ `Operators & execute`

Primary: Division ÷

Keyboard: [/]

#38 1 `Digits & number entry`

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #11 first): D — Variable D (store with STO/ →, recall by name)

Keyboard: [1]

#39 2 `Digits & number entry`

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #11 first): E — Variable E (store with STO/ →, recall by name)

Keyboard: [2]

#40 3 Digits & number entry

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #11 first): F — Variable F (store with STO/ →, recall by name)

Keyboard: 3

#41 + Operators & execute

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #11 first): °′ ″ — Degrees-minutes-seconds entry/display °′ ″

Keyboard: +

#42 – Operators & execute

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #11 first): (–) — Negative sign (–) — makes the next number negative. Different from the subtraction key

Keyboard: -

#43 0 Digits & number entry

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #11 first): x — Variable X (store with STO/ →, recall by name)

Keyboard: 0

#44 • `Digits & number entry`

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #11 first): y — Variable Y (store with STO/ →, recall by name)

Keyboard: `.` `,`

#45 ×10▭ `Digits & number entry`

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

shifted (2nd / f / SHIFT layer) (press #11 first): z — Variable M (store with STO/ →, recall by name)

Keyboard: `e`

#46 FORMAT `Shift, modes, menus & navigation`

Primary: FORMAT — choose number format (Fix/Sci/Norm)

#47 = `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

shifted (2nd / f / SHIFT layer) (press #11 first): ≈ — ▶Decimal — show the result as a decimal

Keyboard: `Enter` `=`

5 · Worked examples

18 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#38 1 → #39 2 → #41 + → #40 3 → #36 × → #33 4 → #47 =

Display shows 24 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#26 (→ #38 1 → #39 2 → #41 + → #29 8 → #27) → #37 ÷ → #33 4 → #47 =

Display shows 5 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#26 (→ #40 3 → #41 + → #33 4 → #27) → #36 × → #26 (→ #30 9 → #42 − → #39 2 → #27) → #47 =

Display shows 49 **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press Reset first.

#38 1 → #39 2 → #20 x^2 → #47 =

Display shows 144 **— expected 144. ✔ verified on the app.**

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate √144. Press Reset first.

#18 √▭ → #38 1 → #33 4 → #33 4 → #47 =

Display shows 12 **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate 2^10. Press Reset first.

#39 2 → #19 $x^{\square}$ → #38 1 → #43 0 → #47 =

Display shows 1024 **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.7 — Reciprocals

Calculate 1 ÷ 0.25 (x^{-1}). Press Reset first.

#43 0 → #44 • → #39 2 → #34 5 → #11 SHIFT → #19 $\square^{-1}$ → #47 =

Display shows 4 **— expected 4. ✔ verified on the app.**

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.8 — Trigonometry (degrees)

Calculate tan 45° (Deg mode). Press Reset first.

#25 tan → #33 4 → #34 5 → #27) → #47 =

Display shows 1 **— expected 1. ✔ verified on the app.**

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#11 SHIFT → #20 log → #38 1 → #43 0 → #43 0 → #43 0 → #27) → #47 =

Display shows 3 — expected 3. ✔ verified on the app.

log is base-10: it counts the zeros of a power of ten.

Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#11 SHIFT → #21 ln → #38 1 → #27) → #47 =

Display shows 0 — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.11 — Negative numbers

Calculate **−5 + 12**. Press **Reset** first.

#11 SHIFT → #42 (−) → #34 5 → #41 + → #38 1 → #39 2 → #47 =

Display shows 7 — expected 7. ✔ verified on the app.

Use the sign key for a negative number — not the subtraction key.

Example 5.12 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

#35 6 → #45 ×10□ → #39 2 → #36 × → #40 3 → #47 =

Display shows 1800 **— expected 1800. ✔ verified on the app.**

The exponent key enters ×10ⁿ without typing × 10 ^.

Example 5.13 — Area of a circle πr²

Calculate π × 2.5². Press Reset first.

#11 SHIFT → #28 π → #36 × → #39 2 → #44 • → #34 5 → #20 x^2 → #47 =

Display shows 19.63495408 **— expected 19.63495408. ✔ verified on the app.**

π is stored to full precision — never type 3.14.

Example 5.14 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#38 1 → #43 0 → #43 0 → #36 × → #33 4 → #47 = → #37 ÷ → #39 2 → #47 =

Display shows 200 **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.15 — Average of three numbers

Calculate **(4 + 8 + 15) ÷ 3**. Press **Reset** first.

#26 (→ #33 4 → #41 + → #29 8 → #41 + → #38 1 →
#34 5 → #27) → #37 ÷ → #40 3 → #47 =

Display shows `9` — expected 9. ✔ verified on the app.

Add up the values, then divide by how many there are.

Example 5.16 — Mixed operations

Calculate **50 − 4 × 6 + 7**. Press **Reset** first.

#34 5 → #43 0 → #42 − → #33 4 → #36 × → #35 6 →
#41 + → #28 7 → #47 =

Display shows `33` — expected 33. ✔ verified on the app.

Only the × part is done first; the − and + are then worked left to right.

Example 5.17 — Dividing twice

Calculate **360 ÷ 4 ÷ 9**. Press **Reset** first.

#40 3 → #35 6 → #43 0 → #37 ÷ → #33 4 → #37 ÷ →
#30 9 → #47 =

Display shows `10` — expected 10. ✔ verified on the app.

Division is worked left to right: the first quotient is divided again.

Example 5.18 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#26 (→ #26 (→ #40 3 → #41 + → #34 5 → #27) → #36 × → #33 4 → #42 − → #39 2 → #27) → #37 ÷ → #38 1 → #43 0 → #47 =

Display shows 3 — expected 3. ✔ verified on the app.

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

29 extra functions sit on shifted layers. Press the layer key once, then the key: for example #11 then #16 → ⇄; #11 then #17 → ▭⌟▭/▭; #11 then #18 → $\sqrt[3]{▭}$; #11 then #19 → $▭^{-1}$.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #47.

Hack 13: Type faster on a computer

27 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
ArrowUp	#8	⌃ ⌄	Cursor ▲ — up / previous history line / previous item
r	#18	√▭	Square root √
^	#19	x▭	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
q	#20	x^2	Square x^2
s	#23	sin	Sine (uses the current angle unit)
c	#24	cos	Cosine (current angle unit)
t	#25	tan	Tangent (current angle unit)
(	#26	(	Opening bracket (
)	#27	)	Closing bracket). Closing brackets at the end of a line may be omitted
7	#28	7	Digit 7
8	#29	8	Digit 8
9	#30	9	Digit 9
Backspace Delete	#31	⌫	DEL — delete the character at the cursor
4	#33	4	Digit 4
5	#34	5	Digit 5
6	#35	6	Digit 6
* x	#36	×	Multiplication ×
/	#37	÷	Division ÷
1	#38	1	Digit 1
2	#39	2	Digit 2

3	#40	3	Digit 3
+	#41	+	Addition +
-	#42	−	Subtraction −
0	#43	0	Digit 0
. ,	#44	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#45	×10▭	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#47	=	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (94)

Work each one on calculator #14, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (250 + 350) ÷ 12 = _____
14. (5 + 6) × (10 − 3) = _____
15. (12 + 3) × (8 − 5) = _____
16. (1.5 + 2.5) × (7 − 4) = _____

17. (20 + 30) × (15 − 5) = _____

18. (9 + 1) × (100 − 1) = _____

19. (11 + 13) × (17 − 9) = _____

20. 25^2 = _____

21. 7^2 = _____

22. 1.5^2 = _____

23. 19^2 = _____

24. 33^2 = _____

25. 0.5^2 = _____

26. 125^2 = _____

27. 64^2 = _____

28. √625 = _____

29. √1024 = _____

30. √9801 = _____

31. 3^5 = _____

32. 1.05^3 = _____

33. 5^4 = _____

34. 7^3 = _____

35. 1.5^2 = _____

36. 10^6 = _____

37. 2^0.5 = _____

38. 1 ÷ 0.2 (x^{-1}) = _____

39. sin 90° (Deg mode) = _____

40. cos 0° (Deg mode) = _____

41. log 100 = _____

42. log 10000 = _____

43. ln 10 = _____

44. ln 2 = _____

45. −20 + 7 = _____

46. −3 + 3 = _____

47. −100 + 250 = _____

48. −8 + 1 = _____

49. 2×10^3 × 4 = _____

50. 1.5×10^4 × 2 = _____

51. 3×10^5 × 3 = _____

52. 4.5×10^2 × 2 = _____

53. π × 1.2^2 = _____

54. 15 × 6 = , then Ans ÷ 9 = _____

55. 7.5 × 8 = , then Ans ÷ 3 = _____

56. 250 × 12 = , then Ans ÷ 100 = _____

57. (12 + 15 + 18) ÷ 3 = _____

58. (70 + 85 + 91) ÷ 3 = _____

59. (2.5 + 3.5 + 6) ÷ 3 = _____

60. (100 + 200 + 600) ÷ 3 = _____

61. 100 − 3 × 12 + 9 = _____

62. 18 − 2 × 5 + 40 = _____

63. 7.5 − 1.5 × 3 + 2 = _____

64. 64 − 8 × 4 + 10 = _____

65. 300 − 25 × 4 + 55 = _____

66. 9 − 3 × 3 + 1 = _____

67. 45 − 5 × 5 + 15 = _____

68. 1000 ÷ 8 ÷ 5 = _____

69. 96 ÷ 2 ÷ 3 = _____

70. 7200 ÷ 60 ÷ 60 = _____

71. 500 ÷ 4 ÷ 25 = _____

72. 81 ÷ 3 ÷ 3 = _____

73. ((10 + 2) × 3 − 6) ÷ 6 = _____

74. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

75. ((20 + 30) × 2 − 25) ÷ 3 = _____

76. ((7 + 8) × 9 − 15) ÷ 12 = _____

77. 12 + 3 × 4 = _____

78. (12 + 8) ÷ 4 = _____

79. (3 + 4) × (9 − 2) = _____

80. 12^2 = _____

81. √144 = _____

82. 2^10 = _____

83. 1 ÷ 0.25 (x^{-1}) = _____

84. tan 45° (Deg mode) = _____

85. log 1000 = _____

86. ln 1 = _____

87. −5 + 12 = _____

88. 6×10^2 × 3 = _____

89. π × 2.5^2 = _____

90. 100 × 4 = , then Ans ÷ 2 = _____

91. (4 + 8 + 15) ÷ 3 = _____

92. 50 − 4 × 6 + 7 = _____

93. 360 ÷ 4 ÷ 9 = _____

94. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (66)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “⏻” (ON)? _____
2. Which key is “⌂” (HOME)? _____
3. Which key is “ ▲ ” (Cursor ▲)? _____
4. Which key is “ ▼ ” (Cursor ▼)? _____
5. Which key is “◀” (Cursor ◀)? _____
6. Which key is “▶” (Cursor ▶)? _____
7. Which key is “OK” (OK)? _____
8. Which key is “ ^ ⌄ ” (Cursor ▲)? _____
9. Which key is “⚙” (Settings)? _____
10. Which key is “↩” (Back ↩)? _____
11. Which key is “SHIFT” (SHIFT / 2nd)? _____
12. Which key is “x▭” (Variable list)? _____
13. Which key is “f(x)” (FUNCTION menu)? _____
14. Which key is “☰” (CATALOG)? _____
15. Which key is “⚒” (TOOLS)? _____
16. Which key is “x” (Variable x for formulas, tables and solving)? _____
17. Which key is “▭/▭” (Fraction template ▭/▭)? _____
18. Which key is “√▭” (Square root √)? _____
19. Which key is “x▭” (Power x^y / ^)? _____
20. Which key is “x^2” (Square x^2)? _____

21. Which key is “log▭▭” (log with any base: $\log_a$)? _____

22. Which key is “Ans” (Ans)? _____

23. Which key is “sin” (Sine)? _____

24. Which key is “cos” (Cosine)? _____

25. Which key is “tan” (Tangent)? _____

26. Which key is “(” (Opening bracket)? _____

27. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

28. Which key is “⌫” (DEL)? _____

29. Which key is “AC” (ON / AC)? _____

30. Which key is “×” (Multiplication ×)? _____

31. Which key is “÷” (Division ÷)? _____

32. Which key is “+” (Addition +)? _____

33. Which key is “−” (Subtraction −)? _____

34. Which key is “•” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

35. Which key is “×10▭” (×10^x / EXP / EE)? _____

36. Which key is “FORMAT” (FORMAT)? _____

37. Which key is “=” (Execute / =)? _____

38. How do you reach “⇄”? _____

39. How do you reach “▭⌟▭/▭”? _____

40. How do you reach "$\sqrt[3]{\square}$"? _____

41. How do you reach "$\square^{-1}$"? _____

42. How do you reach "log"? _____

43. How do you reach "ln"? _____

44. How do you reach "DRG▶"? _____

45. How do you reach "$\sin^{-1}$"? _____

46. How do you reach "$\cos^{-1}$"? _____

47. How do you reach "$\tan^{-1}$"? _____

48. How do you reach "="? _____

49. How do you reach ","? _____

50. How do you reach "π"? _____

51. How do you reach "e"? _____

52. How do you reach "i"? _____

53. How do you reach "INS"? _____

54. How do you reach "OFF"? _____

55. How do you reach "A"? _____

56. How do you reach "B"? _____

57. How do you reach "C"? _____

58. How do you reach "D"? _____

59. How do you reach "E"? _____

60. How do you reach “F”? _____

61. How do you reach “°′ ″”? _____

62. How do you reach “(–)”? _____

63. How do you reach “x”? _____

64. How do you reach “y”? _____

65. How do you reach “z”? _____

66. How do you reach “≈”? _____

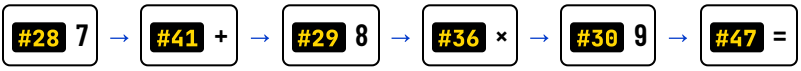
Appendix A — Drill answers

Set A

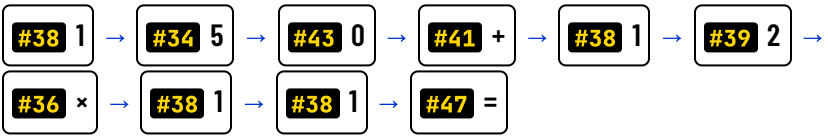
1. 55 (display `55`)



2. 79 (display `79`)



3. 282 (display `282`)



4. 213 (display `213`)



5. 90 (display `90`)



6. 1100 (display `1100`)



7. 82 (display 82)



8. 128 (display 128)



9. 10 (display 10)



10. 8 (display 8)



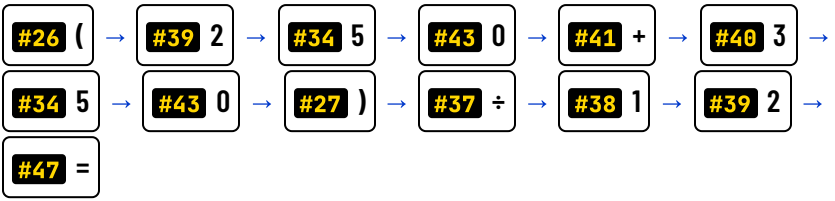
11. 20 (display 20)



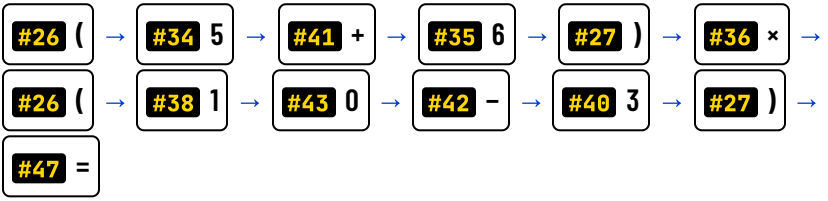
12. 10 (display 10)



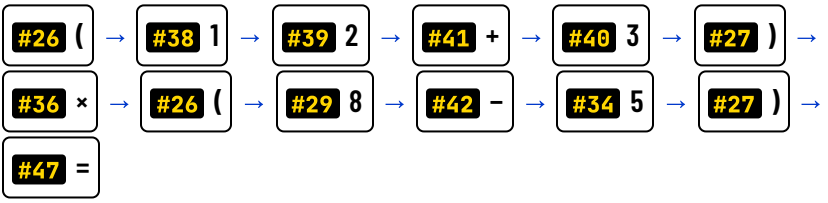
13. 50 (display `50`)



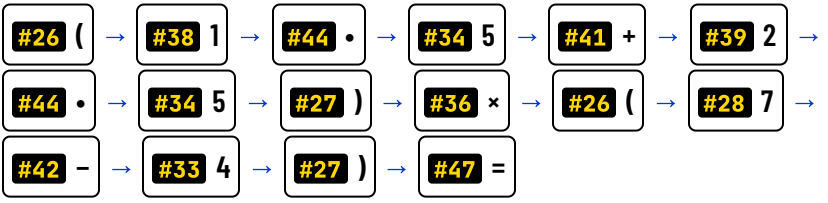
14. 77 (display `77`)



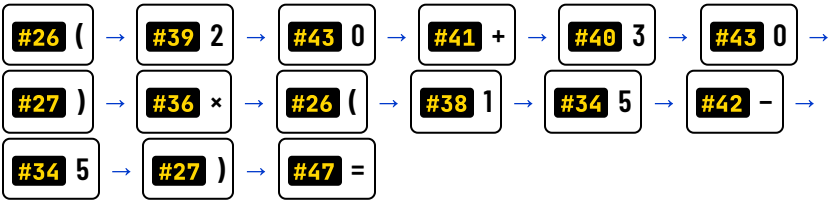
15. 45 (display `45`)



16. 12 (display `12`)



17. 500 (display `500`)



18. 990 (display `990`)

`#26` (→ `#30` 9 → `#41` + → `#38` 1 → `#27`) → `#36` × →
`#26` (→ `#38` 1 → `#43` 0 → `#43` 0 → `#42` − → `#38` 1 →
`#27`) → `#47` =

19. 192 (display `192`)

`#26` (→ `#38` 1 → `#38` 1 → `#41` + → `#38` 1 → `#40` 3 →
`#27`) → `#36` × → `#26` (→ `#38` 1 → `#28` 7 → `#42` − →
`#30` 9 → `#27`) → `#47` =

20. 625 (display `625`)

`#39` 2 → `#34` 5 → `#20` x^2 → `#47` =

21. 49 (display `49`)

`#28` 7 → `#20` x^2 → `#47` =

22. 2.25 (display `2.25`)

`#38` 1 → `#44` • → `#34` 5 → `#20` x^2 → `#47` =

23. 361 (display `361`)

`#38` 1 → `#30` 9 → `#20` x^2 → `#47` =

24. 1089 (display `1089`)

`#40` 3 → `#40` 3 → `#20` x^2 → `#47` =

25. 0.25 (display `0.25`)

`#43` 0 → `#44` • → `#34` 5 → `#20` x^2 → `#47` =

26. 15625 (display `15625`)

`#38` 1 → `#39` 2 → `#34` 5 → `#20` x^2 → `#47` =

27. 4096 (display `4096`)

`#35` 6 → `#33` 4 → `#20` x^2 → `#47` =

28. 25 (display `25`)

`#18` $\sqrt{\square}$ → `#35` 6 → `#39` 2 → `#34` 5 → `#47` =

29. 32 (display `32`)

`#18` $\sqrt{\square}$ → `#38` 1 → `#43` 0 → `#39` 2 → `#33` 4 → `#47` =

30. 99 (display `99`)

`#18` $\sqrt{\square}$ → `#30` 9 → `#29` 8 → `#43` 0 → `#38` 1 → `#47` =

31. 243 (display `243`)

`#40` 3 → `#19` $x^{\square}$ → `#34` 5 → `#47` =

32. 1.157625 (display `1.157625`)

`#38` 1 → `#44` • → `#43` 0 → `#34` 5 → `#19` $x^{\square}$ → `#40` 3 → `#47` =

33. 625 (display `625`)

`#34` 5 → `#19` $x^{\square}$ → `#33` 4 → `#47` =

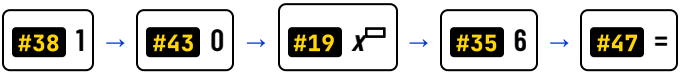
34. 343 (display `343`)

`#28` 7 → `#19` $x^{\square}$ → `#40` 3 → `#47` =

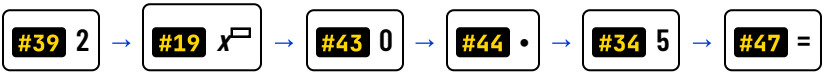
35. 2.25 (display `2.25`)



36. 1000000.000 (display `1000000`)



37. 1.41421356 (display `1.414213562`)



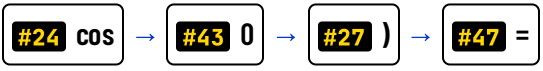
38. 5 (display `5`)



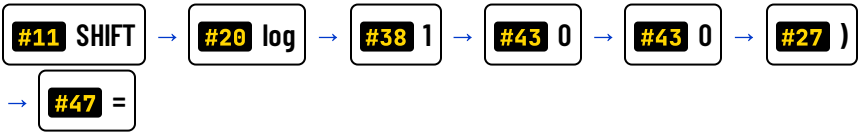
39. 1 (display `1`)



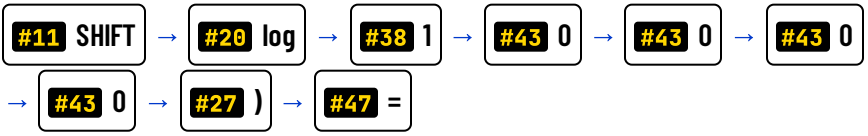
40. 1 (display `1`)



41. 2 (display `2`)



42. 4 (display `4`)



43. 2.30258509 (display `2.302585093`)



44. 0.69314718 (display `0.6931471806`)

#11 SHIFT → #21 ln → #39 2 → #27) → #47 =

45. -13 (display `-13`)

#11 SHIFT → #42 (−) → #39 2 → #43 0 → #41 + → #28 7 → #47 =

46. 0 (display `0`)

#11 SHIFT → #42 (−) → #40 3 → #41 + → #40 3 → #47 =

47. 150 (display `150`)

#11 SHIFT → #42 (−) → #38 1 → #43 0 → #43 0 → #41 + → #39 2 → #34 5 → #43 0 → #47 =

48. -7 (display `-7`)

#11 SHIFT → #42 (−) → #29 8 → #41 + → #38 1 → #47 =

49. 8000 (display `8000`)

#39 2 → #45 ×10□ → #40 3 → #36 × → #33 4 → #47 =

50. 30000 (display `30000`)

#38 1 → #44 • → #34 5 → #45 ×10□ → #33 4 → #36 × → #39 2 → #47 =

51. 900000 (display `900000`)

#40 3 → #45 ×10□ → #34 5 → #36 × → #40 3 → #47 =

52. 900 (display `900`)



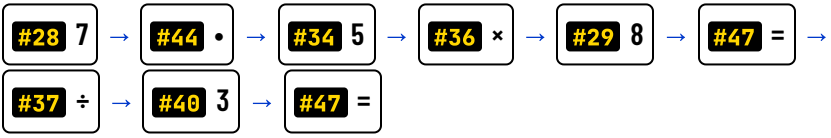
53. 4.52389342 (display `4.523893421`)



54. 10 (display `10`)



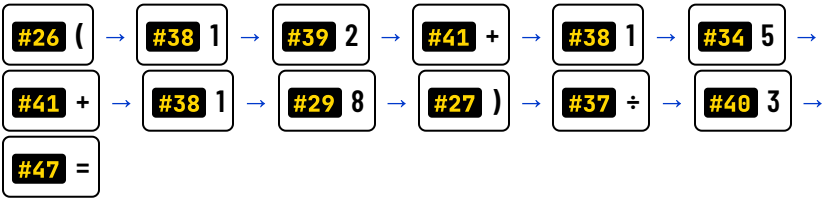
55. 20 (display `20`)



56. 30 (display `30`)



57. 15 (display `15`)



58. 82 (display 82)

#26 (→ #28 7 → #43 0 → #41 + → #29 8 → #34 5 → #41 + → #30 9 → #38 1 → #27) → #37 ÷ → #40 3 → #47 =

59. 4 (display 4)

#26 (→ #39 2 → #44 • → #34 5 → #41 + → #40 3 → #44 • → #34 5 → #41 + → #35 6 → #27) → #37 ÷ → #40 3 → #47 =

60. 300 (display 300)

#26 (→ #38 1 → #43 0 → #43 0 → #41 + → #39 2 → #43 0 → #43 0 → #41 + → #35 6 → #43 0 → #43 0 → #27) → #37 ÷ → #40 3 → #47 =

61. 73 (display 73)

#38 1 → #43 0 → #43 0 → #42 − → #40 3 → #36 × → #38 1 → #39 2 → #41 + → #30 9 → #47 =

62. 48 (display 48)

#38 1 → #29 8 → #42 − → #39 2 → #36 × → #34 5 → #41 + → #33 4 → #43 0 → #47 =

63. 5 (display 5)

#28 7 → #44 • → #34 5 → #42 − → #38 1 → #44 • → #34 5 → #36 × → #40 3 → #41 + → #39 2 → #47 =

64. 42 (display 42)



65. 255 (display 255)



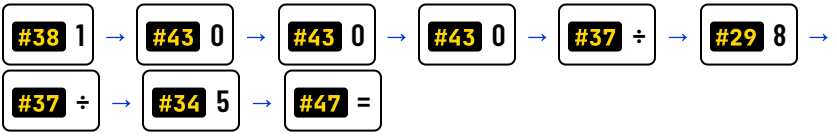
66. 1 (display 1)



67. 35 (display 35)



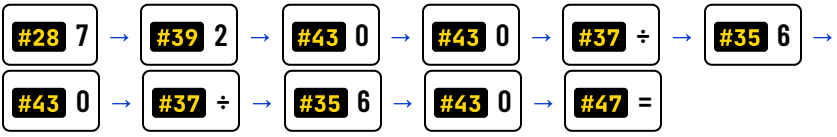
68. 25 (display 25)



69. 16 (display 16)



70. 2 (display 2)



71. 5 (display 5)

#34 5 → #43 0 → #43 0 → #37 ÷ → #33 4 → #37 ÷ → #39 2 → #34 5 → #47 =

72. 9 (display 9)

#29 8 → #38 1 → #37 ÷ → #40 3 → #37 ÷ → #40 3 → #47 =

73. 5 (display 5)

#26 (→ #26 (→ #38 1 → #43 0 → #41 + → #39 2 → #27) → #36 × → #40 3 → #42 − → #35 6 → #27) → #37 ÷ → #35 6 → #47 =

74. 4 (display 4)

#26 (→ #26 (→ #38 1 → #44 • → #34 5 → #41 + → #39 2 → #44 • → #34 5 → #27) → #36 × → #35 6 → #42 − → #33 4 → #27) → #37 ÷ → #34 5 → #47 =

75. 25 (display 25)

#26 (→ #26 (→ #39 2 → #43 0 → #41 + → #40 3 → #43 0 → #27) → #36 × → #39 2 → #42 − → #39 2 → #34 5 → #27) → #37 ÷ → #40 3 → #47 =

76. 10 (display `10`)

`#26` (→ `#26` (→ `#28` 7 → `#41` + → `#29` 8 → `#27`) →
`#36` × → `#30` 9 → `#42` − → `#38` 1 → `#34` 5 → `#27`) →
`#37` ÷ → `#38` 1 → `#39` 2 → `#47` =

77. 24 (display `24`)

`#38` 1 → `#39` 2 → `#41` + → `#40` 3 → `#36` × → `#33` 4 →
`#47` =

78. 5 (display `5`)

`#26` (→ `#38` 1 → `#39` 2 → `#41` + → `#29` 8 → `#27`) →
`#37` ÷ → `#33` 4 → `#47` =

79. 49 (display `49`)

`#26` (→ `#40` 3 → `#41` + → `#33` 4 → `#27`) → `#36` × →
`#26` (→ `#30` 9 → `#42` − → `#39` 2 → `#27`) → `#47` =

80. 144 (display `144`)

`#38` 1 → `#39` 2 → `#20` x^2 → `#47` =

81. 12 (display `12`)

`#18` $\sqrt{\square}$ → `#38` 1 → `#33` 4 → `#33` 4 → `#47` =

82. 1024 (display `1024`)

`#39` 2 → `#19` $x^{\square}$ → `#38` 1 → `#43` 0 → `#47` =

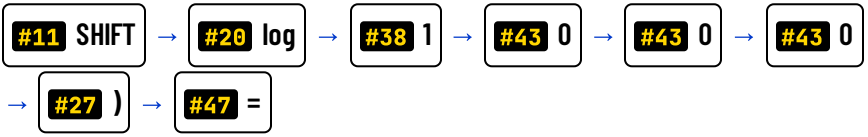
83. 4 (display `4`)



84. 1 (display `1`)



85. 3 (display `3`)



86. 0 (display `0`)



87. 7 (display `7`)



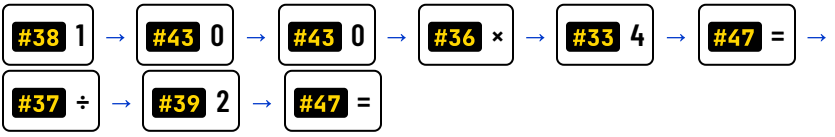
88. 1800 (display `1800`)



89. 19.63495408 (display `19.63495408`)



90. 200 (display `200`)



91. 9 (display 9)

#26 (→ #33 4 → #41 + → #29 8 → #41 + → #38 1 → #34 5 → #27) → #37 ÷ → #40 3 → #47 =

92. 33 (display 33)

#34 5 → #43 0 → #42 − → #33 4 → #36 × → #35 6 → #41 + → #28 7 → #47 =

93. 10 (display 10)

#40 3 → #35 6 → #43 0 → #37 ÷ → #33 4 → #37 ÷ → #30 9 → #47 =

94. 3 (display 3)

#26 (→ #26 (→ #40 3 → #41 + → #34 5 → #27) → #36 × → #33 4 → #42 − → #39 2 → #27) → #37 ÷ → #38 1 → #43 0 → #47 =

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26

27. #27
28. #31
29. #32
30. #36
31. #37
32. #41
33. #42
34. #44
35. #45
36. #46
37. #47
38. #11 SHIFT then #16 x
39. #11 SHIFT then #17 ▭/▭
40. #11 SHIFT then #18 √▭
41. #11 SHIFT then #19 x▭
42. #11 SHIFT then #20 x²
43. #11 SHIFT then #21 log▭▭
44. #11 SHIFT then #22 Ans
45. #11 SHIFT then #23 sin
46. #11 SHIFT then #24 cos
47. #11 SHIFT then #25 tan
48. #11 SHIFT then #26 (
49. #11 SHIFT then #27)
50. #11 SHIFT then #28 7
51. #11 SHIFT then #29 8
52. #11 SHIFT then #30 9
53. #11 SHIFT then #31 ⌫
54. #11 SHIFT then #32 AC
55. #11 SHIFT then #33 4
56. #11 SHIFT then #34 5
57. #11 SHIFT then #35 6
58. #11 SHIFT then #38 1
59. #11 SHIFT then #39 2
60. #11 SHIFT then #40 3
61. #11 SHIFT then #41 +
62. #11 SHIFT then #42 -
63. #11 SHIFT then #43 0
64. #11 SHIFT then #44 •
65. #11 SHIFT then #45 ×10▭
66. #11 SHIFT then #47 =

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23	×10▭	#45
▶Decimal — show the result as a decimal	≈	#11 → #47
Addition +	+	#41
Ans — recall the last answer into the current expression	Ans	#22
Back ↩ — previous screen	↩	#10
CATALOG — list of all commands	☰	#14
Closing bracket). Closing brackets at the end of a line may be omitted	)	#27
Comma / argument separator (between function arguments, list items or coordinates)	,	#11 → #27
Common logarithm $\log_{10}$	log	#11 → #20
Cosine (current angle unit)	cos	#24
Cube root $\sqrt[3]{}$	$\sqrt[3]{▭}$	#11 → #18
Cursor ▲ — up / previous history line / previous item	▲	#3
Cursor ▲ — up / previous history line / previous item	^ ⌄	#8
Cursor ▶ — move right in the line	▶	#6
Cursor ▼ — down / next history line / next item	▼	#4
Cursor ◀ — move left in the line	◀	#5
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	•	#44
Degrees-minutes-seconds entry/display °′ ″	°′ ″	#11 → #41
DEL — delete the character at the cursor	⌫	#31
Digit 0	0	#43

Digit 1	1	#38
Digit 2	2	#39
Digit 3	3	#40
Digit 4	4	#33
Digit 5	5	#34
Digit 6	6	#35
Digit 7	7	#28
Digit 8	8	#29
Digit 9	9	#30
Division ÷	÷	#37
DRG — choose/convert the angle unit (Deg / Rad / Grad)	DRG▶	#11 → #22
Equals — complete the calculation	=	#11 → #26
Euler's number e (2.71828…)	e	#11 → #29
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	=	#47
FORMAT — choose number format (Fix/Sci/Norm)	FORMAT	#46
Fraction template ▭/▭ (numerator over denominator)	▭/▭	#17
FUNCTION menu	f(x)	#13
HOME — return to the main calculation screen	⌂	#2
Imaginary unit i (complex numbers)	i	#11 → #30
INS — insert mode at the cursor	INS	#11 → #31
Inverse cosine $\cos^{-1}$	$\cos^{-1}$	#11 → #24
Inverse sine $\sin^{-1}$	$\sin^{-1}$	#11 → #23
Inverse tangent $\tan^{-1}$	$\tan^{-1}$	#11 → #25

log with any base: $\log_a(b)$	log▭▭	#21
Mixed-number template (whole, numerator, denominator)	▭⌟▭/▭	#11 → #17
Multiplication ×	×	#36
Natural logarithm ln	ln	#11 → #21
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#11 → #42
OFF — power off (state is lost; press ON to restart)	OFF	#11 → #32
OK — confirm the highlighted item	OK	#7
ON — power on	⏻	#1
ON / AC — power on, or clear all	AC	#32
Opening bracket (	(	#26
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	x▭	#19
Reciprocal x^{-1} (1/x)	▭$^{-1}$	#11 → #19
S⇔D — toggle the result between exact (fraction, √, π) and decimal	⇄	#11 → #16
Settings	⚙	#9
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#11
Sine (uses the current angle unit)	sin	#23
Square root √	√▭	#18
Square x^2	x^2	#20
Subtraction −	−	#42
Tangent (current angle unit)	tan	#25
TOOLS — tools menu	⚒	#15
Variable A (store with STO/ →, recall by name)	A	#11 → #33

Variable B (store with STO/ →, recall by name)	B	#11 → #34
Variable C (store with STO/ →, recall by name)	C	#11 → #35
Variable D (store with STO/ →, recall by name)	D	#11 → #38
Variable E (store with STO/ →, recall by name)	E	#11 → #39
Variable F (store with STO/ →, recall by name)	F	#11 → #40
Variable list (store / recall)	x▭	#12
Variable M (store with STO/ →, recall by name)	z	#11 → #45
Variable X (store with STO/ →, recall by name)	x	#11 → #43
Variable x for formulas, tables and solving	x	#16
Variable Y (store with STO/ →, recall by name)	y	#11 → #44
π (3.14159265…) as an exact constant	π	#11 → #28

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

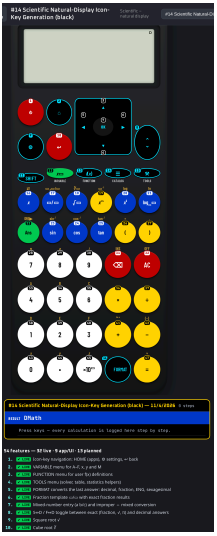
Icon-key generation: HOME opens the calculation apps, ⚙ the settings, VARIABLE / FUNCTION / CATALOG / TOOLS menus, FORMAT converts the last result (decimal, fraction, ENG, sexagesimal). SHIFT + 4/5/6/1/2/3 enter variables A–F; SHIFT 0 / • enter x / y; SHIFT ×10 is mapped to the memory variable M.

Planned (not yet working)

- Spreadsheet app
- Distribution app
- Algorithm (flowchart) app
- Math box: dice / coin simulation
- QR-code export
- Ratio app
- Inequality app
- Prime factorisation
- Exam mode (locks memory & shows indicator)
- Answer history export to text
- Speech read-out of results
- Colour-blind safe key theme
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 ⏻	#2 ⌂	#3 ▲
#4 ▼	#5 ◀	#6 ▶
#7 OK	#8 ^ ⌄	#9 ⚙
#10 ↩	#11 SHIFT	#12 x▭
#13 f(x)	#14 ☰	#15 ⚒
#16 x ⇧ ⇄	#17 ▭/▭ ⇧ ▭⌟▭/▭	#18 √▭ ⇧ ∛▭
#19 x▭ ⇧ ▭$^{-1}$	#20 x^2 ⇧ log	#21 log▭▭ ⇧ ln
#22 Ans ⇧ DRG▶	#23 sin ⇧ sin^{-1}	#24 cos ⇧ cos^{-1}
#25 tan ⇧ tan^{-1}	#26 (⇧ =	#27) ⇧ ,
#28 7 ⇧ π	#29 8 ⇧ e	#30 9 ⇧ i
#31 ⌫ ⇧ INS	#32 AC ⇧ OFF	#33 4 ⇧ A
#34 5 ⇧ B	#35 6 ⇧ C	#36 ×
#37 ÷	#38 1 ⇧ D	#39 2 ⇧ E
#40 3 ⇧ F	#41 + ⇧ °′″	#42 − ⇧ (−)
#43 0 ⇧ x	#44 • ⇧ y	#45 ×10▭ ⇧ z
#46 FORMAT	#47 = ⇧ ≈	

U-45 CALCULATORS · Calculator #14 · cut out and keep